DeltaSieve: Targeted Visual Validation of Spreadsheet Transformations

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***Abstract*—Native visual validation of spreadsheet transformations is expensive when every worksheet must be rendered. DeltaSieve selects potentially affected regions using differences in cell values, effective style properties, layout, and dependencies of conditional formatting and charts. Unknown dependencies retain explicit capture fallbacks. Paired native rebuilds and retained transformation controls distinguish comparison evidence from changes introduced by Excel itself. The system records selected coverage, exact pixel observations, process measurements, and reusable evidence. This working manuscript describes the method and its evaluation protocol, reporting completed bounded checks separately from ongoing corpus observations. Independent full-reference and mutation experiments remain necessary to quantify detection coverage and avoided rendering cost.**

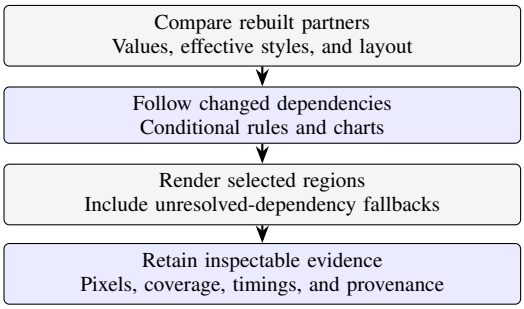


Fig. 1. DeltaSieve separates dependency-aware selection, native observation, and evidence. A proven unchanged dependency can exclude rendering. An unresolved dependency retains its recorded fallback. Targeted coverage remains explicit.

I. Introduction

A spreadsheet transformation can preserve cell values and still change the visible document. Removing an empty formatted cell can expose a background or alter automatic row height. Changing a referenced input can affect a distant conditional-formatting region or chart. Checking only edited cell payloads therefore leaves important transformation effects unobserved.

Native rendering supplies an operational way to observe these effects in the spreadsheet application that users employ. Exhaustive rendering, however, can be expensive for large sheets and corpora. Stored worksheet extents may also include empty or uniformly formatted tails. Dividing these extents by a fixed cell grid can create many images despite very little finite visible content. The challenge is to select useful rendering work while retaining explicit uncertainty about unsupported features and incomplete comparisons.

Figure 1 summarizes the selection and evidence workflow. DeltaSieve combines package-level change analysis with native visual comparison. It follows effective style identities across table renumbering, considers layout and dependency effects, and records the reasons for its capture choices. Supported conditional-formatting and chart dependencies can establish that an unchanged input cannot have changed the associated presentation under the supported model. Changed or unresolved dependencies remain capture candidates. The native stage measures geometry, renders bounded image batches, and compares retained evidence under a recorded state-pair policy.

The scientific question concerns the detection and cost of this selection. A dependency-aware planner is useful only if its reduced work retains the relevant effects in the evaluated domain. We therefore separate selector coverage, pixel observation and defect attribution. Independent full-reference samples and mutations test detection. Repeated same-input observations test rendering variability. A changed image remains an observation whose cause may require further evidence.

Existing spreadsheet dependency slicing motivates careful treatment of indirect effects [1]. Existing workbook comparison tools also provide reports of structural and cell-level changes [2]. DeltaSieve's evaluation must establish the additional contribution of its native-rendering selection and evidence workflow. An interactive Observatory and portable report make those observations inspectable. They support the method's evaluation without replacing its independent detection tests.

DeltaSieve grew out of the need to investigate visual regressions in SheetShrink, a workbook cleanup system described in a companion manuscript [3]. That manuscript studies cleanup rules and preservation structures. Here we study the selection, capture and evidence needed to evaluate rendered changes across workbook transformations, including alternative cleanup tools. The companion working PDF is included in the supplementary material at related-manuscripts/sheetshrink.pdf.

II. Preservation and Comparison Contract

A. Inputs, states, and observations

DeltaSieve validates a pair of spreadsheet states produced by a transformation. The package representation supplies evidence about where visible changes could occur. Native Excel rendering supplies the image observations for selected regions. The two sources of evidence serve different roles. A candidate is a reason to inspect a region; pixel inequality is an observation that still needs attribution.

The fast comparison uses Original and transformed workbooks after a full rebuild and save in their existing, separately owned Excel processes. This gives both partners the native recalculation step. Stored and ordinary-saved XML controls remain available for follow-up. Their changes explain potential causes without automatically enlarging the rebuilt-pair capture plan. An independently requested Check Performance comparison has its own transformed state and coverage.

B. Relevant visual area and candidate coverage

The relevant area includes values, formulas, effective visible formatting, text overflow, and objects. A union across relevant input states is needed to cover content removed by a transformation. Empty trailing geometry has a separate contract. The capture plan must retain enough evidence to distinguish a change inside the relevant area from a larger or smaller exterior canvas.

Candidate selection considers changed cell properties and layout, as well as regions affected through conditional formatting and graphical dependencies. Style and shared-string references require comparison of their effective contents because saving can renumber the corresponding tables. Supported reference reasoning can exclude a dependent region when its rule and inputs are unchanged. A changed dependency makes its affected region a candidate. Unresolved reference semantics require explicit fallback coverage.

A selected-region comparison does not by itself demonstrate that every relevant change in the workbook was selected. We evaluate that property using independently rendered reference areas and controlled mutations. The evaluation must record which reference cases completed and which exceeded their budgets. The paper's coverage claims are bounded by these observations and the supported dependency model.

C. Verdicts, failures, and reusable evidence

A successful comparison requires usable evidence from both partners. Missing hashes, failed native commands, and timeouts remain unsuccessful outcomes. Heartbeats establish add-in reachability and do not establish calculation or capture progress. Durable operation receipts and phase measurements support investigation of stalls.

The raw pixel and geometry observations remain attached to a finding after review. Product regressions, rendering variation, environment-dependent values, and verified capture-extent differences have separate classifications. A volatile formula path explains a possible cell-value change while other changed pixels in the same image can require further analysis. No global pixel tolerance is assumed.

Require: State pair (A, B), observation policy P
Ensure: Scoped comparison evidence and explicit outcomes
1: Record input identities and observation policy
2: Resolve value, effective-style and layout changes
3: Follow supported conditional and chart dependencies
4: Include conservative scopes for unresolved effects
5: Open owned native workers and prepare states under P
6: **for all** bounded packages of selected regions **do**
7: Measure native geometry in both states
8: Record zero-area outcomes and partition visible areas
9: Render paired tiles within pixel and memory budgets
10: Compare complete compatible image receipts
11: Retain differences, equality evidence and failures
12: **end for**
13: Record selected coverage and unfinished work
14: Release owned resources and record confirmed exits

Fig. 2. DeltaSieve's logical comparison stages. Missing or failed captures retain explicit outcomes. Independent reference controls evaluate selection.

Reusable observations bind the input states, relevant configuration, producing validation implementation, and declared rendering environment. Package-content identity is evaluated independently of ZIP entry order and compression. Reusing a completed study preserves its source manifest and compatibility checks. Portable reports retain the observations and their provenance for inspection without rerunning Excel.

III. Method

A. Comparison subject and observation model

Figure 2 separates package analysis, native observation and outcome accounting. The input is an original/transformed pair, a state-preparation policy and a renderer environment. The principal comparison opens separate owned Excel processes and observes both partners after full rebuilding in place. Ordinary-save and pre-rebuild controls retain the effects of serialization and calculation. Conversion of a legacy XLS source precedes this subject and has its own failure outcome.

Let A and B denote the compared states, E their justified relevant visual area and C the selected observation regions. The intended selector obligation is to include every transformation-induced visual difference inside E in C. Pixel equality on C tests the selected observations. It does not establish that the selector met that obligation. Independent reference captures and visible mutations evaluate this second question. Renderer variability and unsupported dependencies remain explicit boundaries of the model.

B. State preparation and attribution controls

Opening a workbook can repair or normalize its stored representation, saving can remove structure, and recalculation can change cached results. The comparison therefore names the operations that produced each observed state. Let W_0 be the raw package, W_s its ordinary-save control and W_r the rebuilt saved control. A native optimization comparison additionally retains W_{cp} after Check Performance and Save As, and W_{cpr} after rebuilding that result. The transformed workbook has its own corresponding controls.

The primary observation compares independently prepared rebuilt partners. Control differences explain which changes arose while opening, saving, optimizing or rebuilding. Ordinary Save and Save As remain distinct generation recipes. One cannot substitute a cheaper recipe under an existing control name and retain its evidence meaning. Legacy format conversion and explicit repair also produce separately identified artifacts.

Each control records its parent identity, native environment and generation recipe. A reusable control artifact saves preparation work but supplies no new visual verdict. A comparison still needs compatible observations of both partners. If opening or a required operation fails, that state remains unavailable. The workflow does not manufacture its image from another state. This separation also permits attributing a reduced package to Excel's save behavior without crediting that reduction to the tested cleanup operation.

C. Streaming change evidence

The package analyzer uses borrowed XML byte views with bounded hierarchy and attribute state. It resolves shared strings, cell formats and relationships before interpreting a coordinate change. String and formula interpretation can require decoding, but worksheet analysis does not instantiate a general XML DOM or an Excel object per cell. This is important on bloated inputs with many empty formatting records.

Cell evidence is written as fixed-size records containing coordinates, a SHA-256 fingerprint and classification flags. The current record is 40 bytes. The record stream resides in an owned scratch file instead of a heap map of cell objects. Sorted streams are merged by coordinate, distinguishing changed, added and removed cells. Equal stream digests avoid repeated cell comparison. The merge is linear in the combined record counts, excluding XML parsing, style resolution, hashing and selected dependency work. Scratch I/O is an explicit cost of this memory organization. Shared strings, styles, layout records, selected changes and native images remain additional memory consumers.

Figure 3 shows the serialized evidence layout. The writer requires unique ascending numeric coordinates, rejects nested or unclosed cells, and hashes a cell's text incrementally. A very large cell body therefore need not become one temporary string. Record readers require complete records. A truncated scratch file cannot become a shorter valid sheet. This strict stream contract permits a two-way merge without a separate sorting pass for supported ordered worksheets.

The fingerprint preserves distinctions that the model has not justified merging. Unsupported serialization can therefore produce conservative extra work rather than a semantic-equivalence claim. Canonicalizing style-table references is a specific supported operation, not a general normalization of arbitrary formulas, XML extensions or renderer behavior. The same byte-processing substrate serves content comparison, supported reference scans and rebuild-eligibility facts, while their decision predicates remain separate.

Keep cell evidence outside the heap

Abbreviated Rust serialization

```
let mut record = [0u8; RECORD_BYTES];
record[..4].copy_from_slice(
    &cell.row.to_le_bytes());
record[4..6].copy_from_slice(
    &cell.column.to_le_bytes());
// Two flag bytes occupy positions 6-7.
record[8..].copy_from_slice(&cell.digest);
output.write_all(&record)?;
```

A 40-byte disk record stores row, column, two flag bytes and a 32-byte digest. Ordered records support a streaming comparison without retaining one heap object per cell.

This is the serialized layout, not the Rust struct layout. Style tables, changed cells and native images still need separate memory.

Fig. 3. The fixed-size cell evidence stream. Coordinates and flags accompany the content/style digest. Abbreviated Rust.

A format index has no cross-package meaning. Style identity resolves fonts, fills, borders, number formats and base-format inheritance, so a permutation of format tables does not create a change at every cell. Changed shared context, including themes or unsupported style representations, retains the conservative path. A removed blank cell can expose an inherited row or column format even when values are unchanged. Conversely, an absent untyped blank and an empty untyped record can supply the same blank input to a conditional rule. The full visual fingerprint and the rule-input fingerprint therefore have different responsibilities.

D. Dependency-aware candidate selection

Direct value, formula, style and layout changes create candidates. Effects propagate beyond their edited coordinates. A row-height change can move later content, a column-width change can move content to its right, and a merged or centered span can extend the required text context. Changed inherited row formats are coalesced into contiguous runs, with unchanged rows splitting the runs. This reduces candidate objects while preserving the union of requested regions.

Conditional-format selection resolves the supported rule formula, application area and referenced inputs. A changed precedent selects the affected application region even when its own appearance is unchanged. Local rules, area-wide inputs and explicitly referenced inputs require different dependency scopes. Exclusion requires unchanged supported rules, their relevant input fingerprints and layout. Unknown names, dynamic references, extensions or unresolved scope retain fallback coverage. DeltaSieve does not evaluate the rule to predict Excel's final color. The native renderer supplies that observation.

Figure 4 shows an unchanged rule whose precedent changes. Selection follows the dependency to its application region. It

Follow the changed input

Abbreviated dependency illustration

```
<!-- Only B1 changes: 0 to 1. -->
<conditionalFormatting sqref="D2:D40">
 <cfRule type="expression" dxfId="0">
  <formula>$B$1&gt;0</formula>
 </cfRule>
</conditionalFormatting>
<!-- The rule is unchanged. -->
<!-- D2:D40 still needs observation. -->
```

Changing B1 can change the rendered fill in D2:D40 even when all cells and rule XML in that area are unchanged. The application area becomes a native capture candidate.

This illustration assumes a supported rule and unchanged differential style. Unknown formulas retain fallback coverage.

Fig. 4. A changed precedent can require observation outside the edited coordinate. Abbreviated XML.

does not infer that unchanged serialized cells inside that region imply unchanged pixels.

E. Invert supported rule references

Enumerating every target cell of a whole-column conditional rule would defeat the purpose of selective rendering. For supported A1 references, DeltaSieve instead maps each changed source coordinate back to dependent target intervals. A relative scalar reference at row a in a rule anchored at row h reads source row $a + t - h$ when applied at target row t. A changed source row x therefore selects target row

$$t = x - a + h.$$

An absolute reference to row a selects all permitted target rows when $x = a$, and none otherwise. Columns follow the corresponding rule. Range endpoints can independently mix absolute and relative axes. The implementation solves the resulting axis intervals, combines row and column intervals and intersects them with the application and comparison extents. It includes both shared and per-area anchor interpretations conservatively for disjoint application areas.

This inversion avoids materializing the entire worksheet grid. Its work still depends on changed inputs, supported reference expressions and application areas. Cross-sheet references require a unique supported source sheet. Dynamic, external or ambiguous references retain fallback. Exhaustive small-grid regression controls compare the inverse result with direct forward reference evaluation, including mixed axes and reversed range endpoints. These tests verify the supported mapping, not arbitrary Excel formula semantics.

Charts have a separate dependency closure. Equal chart XML and cached series values are insufficient because source cells can change independently of the cache. The exclusion

TABLE I
SELECTION OBLIGATIONS. AN EXCLUSION APPLIES ONLY WITHIN THE SUPPORTED REPRESENTATION AND THE RECORDED STATE PAIR.

Effect	Required context
Cell formatting	Resolved cell, row and column components
Row/column layout	Native effective geometry and later content
Conditional rule	Rule, application region and supported input closure
Chart	Source cells, package graph, formatting and live placement
Drawing or OLE	Resolved owner, frame, ink or preview and native bounds
Uniform tail	Complete transitions and a finite relevant envelope

gate requires unchanged chart parts, supported source references, source values and styles, source layout and visibility, conditional dependencies, shared package context and complete equal native placement. Chart sources are inspected beyond the worksheet capture envelope. A source outside the visible table cannot disappear from this dependency check. Missing partners, ambiguous names, user-shape dependencies and unsupported references retain chart capture. A conservative mode exports every chart.

Graphical objects require both package identity and native geometry. Relationship mapping binds an object to its actual worksheet. Supported lines include their stroke envelope. Zero geometric height alone cannot exclude visible ink. Cached previews of embedded objects can render even when Excel labels their shape kind unsupported. Unsupported objects therefore prevent a finite-bound claim unless their own explicit qualification succeeds.

F. Finite observation envelopes

The relevant envelope includes values, formulas, supported visible formatting, text context and qualified object extents from both compared states. Removed content therefore remains in scope. Native-qualified drawing bounds refine the rebuilt-state envelope. Save-control changes retain their own attribution records without automatically enlarging that distinct rebuilt comparison.

An empty exterior row cannot move earlier content. Effective-height selection and image selection use the same lower union boundary, while earlier empty rows remain relevant. Native zero-width and zero-height ranges keep their geometry and a non-rendered outcome. A transition to a visible range remains a finding. No white image is invented for a missing or zero-area observation.

Repeated inherited backgrounds need a finite-bound argument. For supported borderless solid fills, the analyzer retains explicit cells, row and column transitions, contextual neighbors and supported conditional application areas. Uniform unfilled column styles beneath a solid row background introduce no additional horizontal fill edge. Fonts and widths alone do not create such an edge in an empty band. Visible column fills, borders, unknown styles and unsupported rules prevent that exclusion. Supported finite filters and conditional regions

Every live state controls the split

Abbreviated Rust implementation

```
let measured = measure(&batch)?;
ensure!(
    measured.len() == batch.len(),
    "Incomplete geometry batch coverage"
);
for (candidate, geometries) in
    batch.into_iter().zip(measured)
{
    for geometry in geometries.values() {
        geometry.validate()?;
    }
    // Split if any live state is too large.
}
```

The planner measures the same candidate in every live state. A hidden range in one workbook can be visible in its partner. Any oversized state forces further partitioning.

A missing measurement is an error. Zero-area leaves retain an observation. A single oversized cell fails instead of disappearing.

Fig. 5. Every live state participates in geometry qualification. Abbreviated Rust.

remain in the envelope with following context. Full-grid or unresolved application areas retain the conservative path.

These conditions permit less rendering without sampling the selected union. They are relative to the supported model. An unresolved outlying object can still force a full-grid plan and exhaust the available rendering budget.

G. All-state adaptive geometry

Candidate coalescing preserves holes and worksheet identity. Independent ranges are measured in batches through the native adapter. Every live state contributes to image-size qualification. A range hidden in one state can be large in its partner, so any oversized state forces further splitting. Each child is checked again. Large hidden spans can yield zero-area leaves rather than thousands of identical image requests. Figure 5 shows the batch completeness and geometry checks.

The planner respects both area and edge budgets. A conservative point-to-pixel allowance guides partitioning, but returned image dimensions are independently validated. A single cell exceeding the budget fails instead of being omitted. Measured row and column geometry handles hidden or irregular spans that make a fixed cell-count tile unreliable. Later states can repartition an area and backfill earlier observations.

Internal boundaries retain required neighboring context for overflow, merged text and other supported layout effects. The final interior is cropped without resampling. A context region can change rendering even when its own pixels are later cropped. The renderer controls therefore compare a whole

Queue, synchronize, then consume

Abbreviated JavaScript implementation

```
const images = chunk.map(tile =>
  worksheet.getRange(
    CaptureGeometry.address(tile.raw)
  ).getImage()
));
await context.sync();
return images.map(image => image.value);
```

One synchronization completes the queued Office.js image requests. Returned images are then dimension-checked, cropped without resampling and finalized before another bounded batch.

The batch pixel limit also applies. A heartbeat while synchronization is pending does not prove that a capture completed.

Fig. 6. Office.js results are consumed after synchronization. Abbreviated JavaScript.

region with contextual parts and check seam consistency. Agreement of one seam is a local control, not a monolithic oracle for the workbook.

H. Native image qualification

Office.js requests images through `Range.getImage()`. Batching amortizes cross-runtime synchronization. Image results are consumed only after the pending synchronization completes, as shown in Figure 6. The adapter bounds both queued image count and aggregate pixels. Geometry, encoding, transient buffers and upload windows have separate limits.

A successful API response does not prove complete capture. The worker verifies expected dimensions, budget compliance and crop geometry before accepting the image. Returned PNG bytes can first be hashed to avoid uploading identical payloads. Unequal encodings still require decoded-pixel comparison, because PNG serialization is not visual identity. Exact canonical RGBA pixels supply the comparison, with no registration, resampling or general color tolerance. Image receipts bind command, region, state and slot identity, preventing reordered or incomplete upload batches from becoming comparisons of unrelated images.

Raw geometry and origins remain attached to the observation. An exterior-size classification can admit an unpaired right or bottom extension only when each pixel exactly repeats the common boundary. A zero-area counterpart requires an entirely opaque-white retained extent for the corresponding empty-extent classification. These derived classifications retain the original observations. They do not erase changed overlapping content or establish a global tolerance.

I. Attribution after a finding

A native difference can originate in the transformation, calculation, capture context or the environment. DeltaSieve

keeps the observation while diagnostics examine those hypotheses. The formula tracer follows supported A1 references, absolute axes, quoted sheet names, ranges and translated shared formulas from changed saved values. It searches for paths to random/time or environment-dependent functions. Cycles, unsupported expressions and bounded-search exhaustion remain explicit. The search budget limits explanation work, not capture coverage.

A path to a volatile function establishes a dependency, not exclusive causation. Other changed pixels in the same region can have another cause. Source cached values also cannot stand in for values saved from the compared rebuilt states. The report distinguishes their provenance. Repeated original/original controls and unchanged-package controls test renderer behavior independently of the transformation. Counterfactual restoration of a suspected removed witness tests a narrower causal mechanism. None of these operations increases pixel tolerance or automatically clears an unrelated finding.

J. Execution, omission and evidence reuse

Each comparison group owns its Excel processes, private desktop and WebView profile. Office calls are serial within a workbook. Independent groups can progress concurrently under memory admission. Cleanup distinguishes requested termination from confirmed process exit. Setup, opening, geometry, capture, comparison and persistence have separate timing boundaries.

A finite-work watchdog advances after verified work, including validated geometry responses and completed image comparisons. Heartbeats establish reachability while a native call is pending, but do not extend a deadline by themselves. Durable commands and receipts reconcile reconnects. Repeated delivery cannot restart an already completed operation or silently duplicate a pending one. Failed attempts retain produced evidence and receive no equality verdict.

Some observations can be omitted before native work. A direct byte comparison can establish that the original and transformed inputs are identical. ZIP metadata alone does not authorize that omission. A separate rebuild-eligibility check reuses dependency facts from the package scan and a strict part/feature allowlist. Formulas, refresh sources, unknown extensions, duplicate members and unresolved metadata keep rebuilding enabled. Cached values or an empty capture plan do not establish calculation independence.

Completed-result reuse requires compatible source identities, selected parts, implementation, state-preparation policy and rendering environment. Uncompressed member digests make identity independent of ZIP order and compression. A prior part selection must cover every newly requested component. User-selected weaker component policies remain explicit evaluation choices, with correspondingly bounded claims. Interrupted runs and missing image receipts cannot supply completed evidence.

The cache admission condition can be expressed as compatibility of four independent facts. The previous observation must be complete, its required package contents must match, its selected component set must cover the new request, and its renderer/preparation policy must remain compatible. Matching only a filename or producer binary is insufficient. Matching workbook bytes without the renderer contract is likewise insufficient for a native result. A changed compression level can preserve member identity, while a new ordinary save can change it and invalidate admission.

Generation caches, completed-comparison caches and partial working evidence have different roles. An interrupted attempt may retain useful images for inspection, but its missing coverage cannot be supplied by relabeling the attempt complete. Publication records retain the original result and later classification rather than overwriting the evidence after a retry. This is important in a corpus campaign where several machines or attempts contribute observations.

Portable reports preserve input identities, observation regions, exclusions, raw findings, attribution and resource measurements. The Observatory supports inspection and replay without becoming a source of visual truth. Detailed transport schemas, UI controls, experiment launchers and historical failed attempts belong to the versioned artifact. They do not change the selection or pixel-observation contract described here.

K. Machine-Accessible Campaign Monitoring

The source inventories described in Section IV motivate monitoring across long-running campaigns. They contain 34,065 content-addressed Enron table candidates, 9,117 Data.gov XLSX files and 1,182 Destatis XLSX files. Candidate inventory size, successful preparation and completed comparison coverage remain separate quantities.

The control room exposes a local HTTP/JSON interface for compact run summaries, worker and workbook state, comparison reports and retained evidence. Bounded summaries preserve active attempts while limiting completed-item detail. Run and attempt identities prevent observations from earlier executions from being reported as current work. Process liveness, heartbeats and the time of last verified progress are represented separately, allowing an external monitor to distinguish a reachable process from a progressing attempt. A WebSocket stream provides bounded snapshots and incremental status updates for live clients.

These interfaces supported local agent workflows during overnight runs. Agents could inspect stalled attempts and record suspected visual differences for subsequent author review. Capture receipts, pixel comparisons and retained images supply the validation evidence. Agent annotations do not establish visual equality or convert incomplete coverage into a completed result. Explicit control requests and command-status queries support intervention without requiring an agent to infer state from the rendered dashboard.

IV. Evaluation

RQ1 asks whether the selector captures the known visible effects in the tested mutation classes and independent reference

TABLE II
FROZEN XLSX SOURCE INVENTORIES. SIZES ARE DECIMAL GB OF LISTED WORKBOOK FILES AND DO NOT MEASURE COMPLETED NATIVE COVERAGE.

Corpus	XLSX members	XLSX total (GB)
Data.gov	9,117	25.273
Destatis	1,182	2.364

sample. RQ2 measures how much native capture work and elapsed time it avoids. RQ3 examines the reproducibility and investigability of retained findings, including failures and rendering variation.

A. Corpus identity and outcome accounting

The evaluation treats Data.gov, Destatis and Enron as distinct populations. Archive checksums, member names and duplicate occurrences identify each selected corpus version. Legacy XLS conversion is reported separately from XLSX cleanup. A historical Enron development selection contains 20,872 inputs, with 20,503 prepared artifacts and 369 unsuccessful preparation outcomes at its recorded preparation checkpoint. Its selected visual scope contains 16,799 workbooks. These counts describe a development selection and do not establish completed visual coverage. The replacement EDRM Enron v2 XML handover contains 34,065 distinct content-addressed candidates totaling 8.470 decimal GB of uncompressed payload. The imported selection unit is a distinct candidate SHA-256. Candidate transport was checked against the supplied manifests, and the import retained the original candidate bytes. This externally generated extraction includes legacy Excel and other OLE candidates. Native format recognition, successful XLSX conversion and visual eligibility remain separate evaluation stages. Candidate count therefore describes the imported input frame, with workbook outcomes to be reported after processing. The upstream extraction provenance and local import identities are retained alongside the experiment.

Microsoft Graph OLE objects require an additional eligibility distinction. Optional native PowerPoint conversion can recover their chart data into newly generated XLSX workbooks. Those workbooks have a different preparation history from successfully opened original Excel files and are reported as a separate derived-data population. Successful recovery or equality of cell values does not establish visual equivalence of the source Graph chart or coverage of its parent presentation. They do not contribute visual-equality observations without a separately executed native comparison.

The two frozen XLSX inventories in Table II come from the versioned source profiles. Each listed archive member has an individual SHA-256 identity. Their source-frame sizes are separate from the eligible, completed and independently referenced subsets.

A dated evaluation cutoff will report compared workbooks, differences, failures, unsupported representations and unfinished work. Repeated attempts have their own counts. An unchanged package can bypass visual work under an exact input-identity check, but a failed capture cannot contribute an equality observation. The full recursive Enron XML-derived corpus is a separately identified population.

B. Sampling and reference controls

The random evaluation sample is selected from a frozen eligible frame with a recorded seed before inspecting visual findings. Strata use corpus identity, package size and structural features available before validation. Low-reduction and unchanged files remain represented. A separately labelled development stress set contains large and historically problematic workbooks.

Feasible sampled files receive full native reference captures over the relevant visual area using a capture plan independent of the candidate selector. A full-reference timeout remains in the denominator. It is not replaced silently by an easier workbook. Mutations with known visible effects cover cell values, styles, geometry, conditional dependencies and objects. Mutations that Excel normalizes away are reported separately.

A reconciled historical Enron development sample retained twelve seeded source occurrences. Seven completed native references over 33 declared rectangles, two failed image-size preflight and three required unsupported graphical or embedded-object envelopes. All 33 completed rectangles had identical overlapping RGBA pixels. Two had additional opaque-white exterior strips, totaling 196,657 pixels, which were retained as raw extent differences. This sample is not held out, and failed or unsupported references remain in its denominator.

A separate twelve-occurrence Destatis sample used the same fixed-seed, size-quartile selection policy. All twelve selected packages contain graphical or embedded parts and remain unsupported by the current dimension-only reference preparer. They were not replaced by easier inputs. An optional native-object qualification extension admitted one of the same twelve inputs over two fixed areas. Its image rectangle fit inside the cover-sheet area, and both reference areas had identical overlapping RGBA pixels. The data-sheet image had a 20-pixel-high opaque-white exterior strip totaling 25,820 pixels. The remaining eleven inputs retained unsupported outcomes for legacy drawings, embedded objects or drawing effects. This extension checks native object rectangles, with explicit rejection of unsupported types and out-of-area objects. It does not certify decorative ink beyond those rectangles or establish corpus recall.

C. Detection and cost metrics

A completed independent reference determines whether a mutation has a visible effect under the recorded observation policy. A selector miss occurs when such an effect falls outside selected coverage. Detection is measured per mutated workbook pair and per mutation class. A partially covered effect retains a separate outcome even when another difference makes the pair detectable. Pixel coverage requires a documented mapping between reference and selected capture coordinates, including extent differences.

Selection overhead counts candidates whose completed paired images are equal. This measures avoidable work within the tested policy. Pixel findings retain their attribution as transformation effects, rendering variation or unresolved differences. Failed references, missing partner images and timeouts retain explicit outcomes. Report both completed-reference detection and the fraction of all selected cases with usable reference evidence.

D. Mutation strata and statistical scope

The mutation matrix covers direct values and formulas, effective styles, conditional dependencies, chart inputs and objects, row/column geometry and visibility, and unresolved references. Serialization-only controls exercise equivalent package representations. Supported cases and fallback cases have separate expectations. Workbook identities, generator seeds, intended edits, reference areas and observed native effects identify every case. Spreadsheet errors such as division by zero are valid calculation outcomes and need their own fixtures alongside malformed or unopenable packages.

Repeated identical-input native controls characterize renderer variation. Volatile-function cases include NOW and RAND, with INDIRECT and OFFSET also requiring reference-policy controls [4]. The full rebuild and save policy applies to both states. An unchanged formula can therefore produce different rebuilt values. Such observations retain their raw pixels and calculation context for attribution.

Report sample counts and binomial uncertainty for independently sampled units within each declared population [5]. For zero misses in n independent trials with common miss probability, the exact one-sided 95% upper bound is $1 - 0.05^{1/n}$. This is approximately 9.5% for 30 trials and 5.8% for 50. Multiple mutations of one workbook share a cluster and do not establish that many independent workbook observations. Development stress cases remain separately labelled. Population estimates must account for the sampling strata and inclusion probabilities.

E. Matched cost accounting

Cost measurements separate XML planning, geometry, native rendering, transfer, image comparison and persistence where instrumentation permits. Report image count, rendered pixels in each state, compared pixels and retained bytes alongside elapsed time. Overlapping captures count as repeated rendering work; coverage-union size is reported separately. Matched selective and independent reference runs use the same state-preparation policy, environment, image budget and reuse configuration. Analysis, opening, rebuild and save remain part of the declared end-to-end timing even when capture-phase timings are also reported. Batch and concurrency experiments use matched inputs and repeated runs. A changing production queue does not identify the causal benefit of a tuning parameter. Matched timing summaries report paired distributions and repeated-run variation. Full-reference timeouts remain censored observations alongside completion counts. Full-corpus cost estimates use measured visible geometry and stratified samples, with uncertainty and assumptions about hidden regions and object extents.

F. Opening, batching and independent Excel pairs

A bounded opening experiment used four copies of a synthetic workbook with eight million-row, three-column sheets on one Windows host. Two reversed-order rounds measured median batch times of 118.388 seconds sequentially and 41.714 seconds in independent processes sharing a private desktop [6]. The endpoint includes Office.js readiness. Staging and teardown are excluded, and cache and background-load effects remain. This supports overlapping startup for that configuration, not arbitrary concurrent COM calls.

A warmed-up 64-range geometry experiment returned identical measurements in scalar and batched modes. The scalar passes each took 16.094 seconds and the batch passes 0.250 seconds. Twelve-region image packages, including four zero-area regions, took 3.125 seconds at requested size one and 0.579 seconds at size eight, with identical decoded image dimensions and pixels. Opening was excluded and 250-ms observation polling quantizes these timings [7]. The experiment establishes a transport-amortization opportunity under its recorded workload rather than a corpus throughput factor.

Earlier independent-capture trials repeatedly lost one concurrent add-in to its 120-second startup deadline. A later paired experiment used one host and shared inactive desktop with independently owned pair states. Over three repetitions per completed configuration, moving from one pair to two increased capture-only throughput from 3.752 to 7.030 pairs/minute with binary PNG transport and from 7.590 to 12.312 with hash-first transport. Peak observed job memory rose from 1,525 to 2,685 MiB and from 1,472 to 2,666 MiB, respectively. Every tile in this selection was equal, favoring the hash-first path. Requested configurations for three through eight pairs were rejected before launch by RAM admission. Those rejections are not zero-throughput measurements or an Excel scaling ceiling [8], [9].

The historical image-budget protocol selected the smallest budget within 5% of the fastest eligible sum of per-scope median workloads. Both 18-case trials selected 32 MP. The replication sums were 245.14, 235.70 and 230.59 seconds for 16, 32 and 64 MP. Thus 64 MP was fastest and had no observed budget-block failures. Historical local-host, VBA/GDI+ and VSTO/Excel-DNA probes answer separate transport, rasterization or geometry questions. They do not rank interchangeable screenshot engines [10], [11].

G. Coverage qualification before speed

A synthetic million-row request returned an Office.js PNG without an API error in 2.58 seconds. It contained only $85 \times 32{,}766$ pixels against the required 20,971,521-pixel height. The boundary probe found 1,638 rows complete and 1,639 incomplete on that installation [12]. This demonstrates why completion and dimensional qualification are separate conditions. An original-only one-column Lysimeter sweep

subsequently covered its requested 978,624 rows in 576 images and 814.6 seconds after readiness. It supplied hashes and dimensions, not an original/transformed visual verdict.

The historical 62-case protocol completed 48 cases and retained 14 failures. Four unresolved drawing-bound worksheet outcomes account for repeated failures. They are not 14 independent crashes. Six of eight coverage workbooks completed, covering 462 of 466 requested worksheets. The failed observations remained incomplete instead of receiving a smaller undocumented range or an equality classification [10].

H. Reducing work while retaining the selected union

The FRPP workbook has 307,920 rows, including 304,140 hidden rows. Of 19,248 original grid requests, 18,872 have zero area. A read-only geometry probe drove the Rust adaptive planner to 64 leaves after 710 probes in 2.815 seconds. This is a planner observation, not a completed paired Office.js comparison. A separate native visibility-transition control exposed a row only in the transformed state. It retained the original's zero-area observation and reported rendering and geometry differences without claiming paired-pixel comparison there [13].

A 208-worksheet diagnostic exposed 5,192,711 intermediate candidates. Grouping contiguous changed row formats reduced that count to 603,267 with identical content envelopes. Both diagnostics stopped before complete rasterization. This establishes reduced intermediate representation size and no measured end-to-end speedup. It motivates representing shared layout effects as intervals.

Uniform inherited backgrounds exposed a related native-work amplification. One Destatis pair had 128,912 versus 2,197 column definitions. The former plan measured 16,101 regions, although most were outside the explicit-cell envelopes. The guarded finite-background plan produced 165 image pairs, all identical in a fresh comparison. Selection and transport took 41.371 seconds and comparison 1.379 seconds. The earlier attempt was interrupted after more than 40 minutes. A second Enron pair reduced a former 65,664-region plan to 73 measured regions, all with identical images. Its selection and transport took 28.114 seconds. The latter case retained an empty finite AutoFilter and all serialized-cell extent. These are selected development observations with censored earlier attempts, not matched speedup estimates. Regressions retain distant fill transitions, borders, overrides and unsupported rule contexts.

The same Hospital Data original illustrates why candidate counts alone cannot predict rasterization work. On its Financial and Utilization Data sheet, 904 pre-grid candidates yielded four NXPowerLite regions, whereas 905 yielded 3,297 Inquire regions. Inquire also selected 256 regions on another sheet. Selection and transport took 1.921 and 1,296.010 seconds, respectively. Neither comparison found a difference within its selected scope. The two transformed inputs and region sets differ, so these times describe distinct validator workloads.

Conversely, unsupported objects can dominate a conservative plan. Two retained attempts contained explanatory sheets responsible for 104,833 and 185,856 regions. After approximately 5.3 hours, neither full capture had completed. Removing two zero-height lines in a separately supplied reference area changed 1,042 RGBA pixels because their strokes remained visible. The long-running fallback therefore cannot be removed merely because an object's geometric height is zero.

I. Comparator pilot with independent native controls

Five frozen synthetic pairs were supplied to xlCompare 12.5.0.0 and DiffEngineX 3.18.0.0 through their report commands. Both exports included an offscreen B2000 edit. Neither listed a hidden B3000 edit under the tested commands. Direct visual and sparse-formatting changes produced no exported cell differences. A dependent pair exported the changed B1 precedent without a separate conditional-fill or chart-rendering finding. Edition and option coverage remains incomplete, including xlCompare's reported first-ten-rows export limit.

DeltaSieve completed the same five pairs using their ten hash-verified inputs. The identical pair was excluded by byte identity. Four pairs produced native pixel findings. The offscreen edit changed 47 overlapping pixels. Direct, dependent and sparse-formatting pairs changed 7,057, 3,950 and 1,281 grid pixels, respectively. The dependent pair also changed 36,505 chart pixels. The region containing hidden B3000 rendered equally. These observations distinguish edited cell inputs from their dependent visual effects.

An independently supplied reference plan bypassed candidate selection and content clipping. It used Data!A1:N30 for the short pairs, and continuous 256-row tiles through rows 3072 and 2304 for offscreen and sparse pairs. All five comparisons completed. The identical control remained equal. Only the B2000 and J2099 tiles changed in their respective long pairs, while the direct and dependent pairs also differed in their broader reference areas. Pair-level outcomes agree with the selected run. This is a bounded control, not an estimate of corpus recall or pixel-union coverage. Different chart and capture extents prevent pooling their raw pixel counts. No matched cross-tool speedup follows from these trials.

Two additional comparator pairs contained 107 and 215 worksheets. The observed xlCompare times were 53.748 and 21.711 seconds. DiffEngineX required 131.692 and 161.221 seconds and represented every worksheet in its formula-mode reports. These single observations establish execution feasibility and report scope. No matched DeltaSieve cost trial was completed for those pairs.

J. Chart dependency controls

A workbook created in Excel supplies three controlled chart cases. The unchanged pair differs only in document-description metadata. Deterministic selection excludes its one chart, while conservative mode renders both partners. A third pair changes a source value and retains chart capture. Table III records the observed outcomes [14].

These controls test both sides of the exclusion gate. Equal caches alone are not the sufficient condition. The package

TABLE III
NATIVE CHART CONTROLS. IMAGE COUNT INCLUDES BOTH STATES. AN EXCLUDED CHART SUPPLIES NO NEW PAIRED-PIXEL OBSERVATION.

Control	Images	Exclusions	Differences
Unchanged, deterministic	0	1	0
Unchanged, conservative	2	0	0
Changed source, deterministic	2	0	1

and source-cell dependencies and native geometry must also agree. A separate rebuilt Lysimeter inventory contains 69 charts. Its gate excludes 34 and retains 35. Eighteen fallbacks have ambiguous duplicate names and 17 reference additional user-shape dependencies. This inventory did not complete a replacement image comparison and supplies no end-to-end runtime reduction.

K. Attribution controls and retained counterexamples

Independent controls distinguish a transformation defect from capture or renderer behavior. In a fixed-height statistics sheet, the original-versus-original reference was pixel-identical, while the transformed pair repeatedly changed 25,640 pixels. Restoring 87 removed font-bearing blanks recovered the original image hash. A corrected SheetShrink output also matched over the frozen A1:L916 reference area. The case demonstrates a real witness omission rather than an excuse to loosen pixel comparison.

A migration control instead exposed inconsistent exterior context. Its relevant content ended at row 84, but capture included an empty row 85. That row changed from 12.75 points to hidden. A zero-width region then produced a height finding without an image. Applying the same union-content boundary to geometry and capture removed this exterior-context finding. A fresh run completed 38 regions, including 37 pixel comparisons, with no capture-region difference there. Separate height observations remain recorded [15].

Identical-content Enron controls changed only a ZIP comment between their inputs and forced conservative chart capture. Two controls reproduced two and seven changed chart pixels despite unchanged rebuilt worksheet, style and chart parts and equal geometry. They demonstrate native rendering variability without SheetShrink. They justify neither a global color tolerance nor automatic dismissal of another chart's differences. Raw findings and their controls remain separate evidence.

Embedded-object mutations test another boundary. Moving a PDF-icon frame changed 6,473 RGBA pixels in a fixed reference area. Changing only its cached preview changed 5,242. Separately enlarging three zero-size text shapes produced 1,531, 1,737 and 1,487 changed pixels on three sheets, while removal controls were pixel-identical in their supplied areas. These cases show why package anchors, previews and visible ink need distinct treatment. They do not establish coverage for arbitrary OLE formats.

TABLE IV
ENRON STAGE COVERAGE ACROSS NINE SELECTED WORKBOOKS PER TOOL. RAW AND REPAIRED COUNT COMPLETED COMPARISONS BY BASELINE.

Tool	Raw	Repaired	Unavailable	Failed
SheetShrink	8	1	0	0
Check Performance	8	0	1	0
Inquire	7	0	1	1
XLStylesTool	8	0	0	1
NXPowerLite	6	0	2	1
Total	37	1	4	3

L. What the measurements establish

Package bytes, candidate counts, native rectangles, rendered pixels and compared pixels are different units. A reduction in intermediate candidates can lower planner memory without changing the native image count. Adaptive partitioning can reduce images over hidden rows while preserving the same selected cell union. Hash-first transport can reduce transmitted PNG bytes without avoiding Excel's image generation. The experiment ledger records these stages separately.

Similarly, process memory is not a single interchangeable counter. Native Excel, WebView2, the local host and the analysis process can hold different buffers. A sampled private-byte maximum can miss a short allocation peak, and a working-set maximum is not private commit. Scope and sampling interval accompany the reported memory. Phase durations can overlap and are not added as if they were independent. A claim about total campaign throughput includes startup, preparation and failed attempt costs under a declared reuse policy.

The independent reference plan remains distinct from those optimizations. It renders externally supplied areas without candidate selection, conditional-rule exclusion or production content clipping. Its author is responsible for the asserted coverage of those areas. A failed or oversized reference remains a failed reference case. Agreement between a selected plan and a reference over five deliberately constructed pairs supports those controls. It does not turn the same five pairs into a representative sample of three corpora.

M. Recorded coverage of the Enron comparison stage

The first stage uses the nine Enron workbooks with the largest absolute SheetShrink file-size reductions in the frozen selection. Each workbook contributes one comparison slot per tool. Table IV counts completed comparisons under the recorded coverage limits and keeps unavailable variants and failed comparisons in the denominator. The versioned ledger records the outcomes and baseline identities [16].

Two failures followed Excel repair prompts and one required unresolved whole-grid coverage. The repaired Check Performance preparation and the completed SheetShrink comparison at rank two use different baseline hashes. The repaired preparation therefore provides no matched visual observation for that tool slot. These counts describe coverage of the selected high-reduction workbooks. Detection recall requires the independent reference experiments.

N. A border difference inside the common capture

A developer-selected Enron control rendered `Tables!A1:I157` after inspecting a Check Performance finding. Four independent original captures were pixel-identical. SheetShrink and XLStylesTool preserved that reference exactly. Check Performance differed at 126,404 common pixels, while Inquire differed at 490 near the lower edge. Extending the reference through row 160 retained 127,256 and 916 changed common pixels, respectively. Inspection of the extended Inquire capture showed a missing lower horizontal table border inside the shared rectangle. Its rebuilt worksheet lacked original row 157 and its styled blank cells. The added context therefore separates this finding from a capture clipped at the table edge. NXPowerLite had no available output for this input. This post-hoc illustration covers one declared area and does not measure recall or establish whole-workbook preservation. The matching XLStylesTool result is retained alongside SheetShrink's result.

a) Different capture coverage for the same original: For one Data.gov PFAS workbook, the original package-member inventory matched across three tool comparisons. Inquire yielded 129 differing regions among 4,277 selected regions. An inspected range contained purple original pixels that were white in the candidate inside the common image extent. Check Performance yielded no differences in its 63 regions. XLStylesTool selected zero regions. These observations have different denominators and cannot rank the tools' visual preservation. The recorded selection and transport phases were 1,668.137, 24.025 and 0.032 seconds, respectively, and describe the validator's work on those selections.

O. Geometry without a visible surface

Nine zero-width `LCCA` ranges differed in measured height (2000.25 versus 1827 points) without images. Version 16 retains their geometry but excludes these changes from visual findings after validating both zero surfaces. Missing images and visibility transitions remain independently checked. An unchanged-input native repeat completed 29 worksheet and 27 chart observations in 216 seconds with zero visual differences and nine nonvisual geometry observations. Coverage remains limited to the selected areas.

P. Operational outcomes and evaluation limits

A Check Performance preparation campaign attempted 1,182 Destatis inputs and retained 1,016 outputs and 166 failures. Recovery diagnostics identified Name Conflict dialogs for 165 former opening timeouts, while one automation-property failure recurred. A separately retained manual success on another workbook shows that an automation failure does not establish tool incapability. These preparation outcomes remain separate from completed visual observations.

Resource ownership and progress accounting also affect achieved coverage. A peer taskpane restart reproduced under shared-profile cleanup, whereas isolated profiles passed the corresponding bounded controls. In a long geometry phase, verified responses initially failed to advance the watchdog; a corrected repeat completed 741 equal image pairs. These controls motivate isolated resources and finite-work progress without attributing every startup delay to one cause. Historical failure receipts are retained alongside later successful attempts.

The current evidence consists of bounded controls, selected stress cases and partial campaigns. It does not establish population-wide recall, corpus-wide visual equivalence or a matched end-to-end speedup. Independent seeded reference samples, the completed visible-mutation matrix and common-endpoint cost trials remain the empirical obligations. Complete, different, unsupported, failed and unfinished outcomes retain distinct denominators. Supplementary manifests and source-linked reports retain exact input names, revisions, settings and repeated attempts without making their chronology part of the method.

V. Related Work

A. Spreadsheet change inference

SheetDiff infers changes between spreadsheet versions and presents them in a single contextual view [17]. This establishes spreadsheet change inference as prior work. RowColAlign extends longest-common-subsequence reasoning to two-dimensional alignment and evaluates edit recovery using generated pairs with planted changes [18]. Its known edit history provides a methodological precedent for controlled mutations. Our evaluation additionally requires native observations of which mutations produce visible effects under the chosen state-preparation policy. Edit recovery and rendered-effect coverage have different observation units.

B. Spreadsheet dependencies and testing

Spreadsheet slicing connects dependency analysis with integrated testing and fault localization [1]. DeltaSieve uses supported dependencies to select native observation regions. The two methods have different adequacy obligations. Formula-dependency coverage alone does not cover every visual effect.

C. Workbook comparison and native evidence

Spreadsheet Compare documents workbook comparison reports with formula and cell-format categories, including hidden sheets [2]. This is an important existing capability. DeltaSieve's comparison must examine native rendered effects, candidate selection and corpus execution explicitly, using common test inputs where the installed tools support them.

xlCompare documents command-line comparison and exported difference reports [19]. DiffEngineX documents whole-workbook and folder comparisons, report workbooks, and comparisons of formulas and constants. Its help explicitly excludes chart comparison [20]. These documented capabilities motivate a shared input pilot with exported reports. A report of edited cell inputs and an observation of their dependent rendered consequences answer different questions. The pilot must therefore record the tested options and report coverage instead of treating every comparator as an interchangeable visual oracle.

Synkronizer's edition documentation lists content, formatting and Excel-name comparison features [21]. Spreadsheets-Diff advertises two- and three-way comparison, key-column matching, and workbook/VBA comparison features [22]. These vendor descriptions identify further comparison candidates. A native pixel-observation contract and its tested coverage require separate verification for each product and version.

D. Cross-system testing and parsing

SQuaLity examines barriers to reusing database test suites across systems, including configuration and execution dependencies [23]. This suggests recording the renderer and environment when reusing observations. The analogy does not transfer database test adequacy guarantees to spreadsheet rendering. SheetReader's specialized parsing work [24] is relevant to efficient package inspection, whose cost must be separated from native rendering cost.

E. Selective regression testing and visual models

Ekstazi selects regression tests through recorded file dependencies [25]. It tracks dynamically used files and their checksums. Its separation of analysis, execution and dependency collection motivates measuring selection overhead together with avoided work. DeltaSieve's spreadsheet dependency model and native image observations require their own coverage evaluation.

ReDeCheck compares responsive web layout models and reports diagnostic screenshots [26]. It samples viewport widths and refines layout transitions. DeltaSieve instead follows workbook-state changes and supported spreadsheet dependencies. Both settings motivate distinguishing a model-level warning, an observed visual effect and a confirmed defect. Their models and validation subjects require separate experimental oracles.

VI. Discussion and Limitations

The central risk is an incomplete candidate model. A proof for supported inputs must state its dependencies and applicable namespace or rule subset. Unknown features need a capture fallback or an explicitly incomplete outcome. Runtime limits and failures reduce achieved coverage and must remain in the report.

A native renderer is also an experimental dependency. Recalculation, volatile formulas, file-location functions and rendering variability can change output without a transformation defect. A volatile dependency trace explains a supported path rather than every image difference. A white exterior canvas difference requires geometry and containment checks before classification. No global pixel tolerance is inferred from one rare anti-aliasing case.

The Observatory reduces the effort of investigating these outcomes. Portable reports and a future recorded-event replay can demonstrate execution and make findings inspectable. They do not establish authenticity solely by being detailed, and replay cannot supply missing historical telemetry. Scientific claims rely on source identities, implementation versions and the independent evaluation.

VII. Conclusion

DeltaSieve organizes targeted native spreadsheet comparison around inspectable change evidence and explicit coverage. The remaining empirical obligation is to measure detection against independent native references and to quantify cost on matched inputs. The artifact retains observations and uncertainty so those claims can be checked at a fixed reporting cutoff.

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The author directed development, reviewed source code, and performed manual testing and debugging. Agent observations supported investigation. Reported validation outcomes are grounded in the retained native captures and comparison records. The author remains responsible for the scientific claims, references, evidence selection, originality and final submitted content.

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